Latency :-

↳ also known as delay..

Latency is the time taken for data to travel from the client to the server & server to the client

⇒ In simply term, it is delay in communication.

Low latency ↓ ↑ faster response time.

Eg

If you send a request to a server & it responds comes in 1000 Nanosecs then latency 1000 Nanos.

{Maximum latency will been said in ms}.

Realworld

when you click on submit / any button which perform a another to load a new so that the Time taken ⇔ Latency..
to load

The latency defines how long it takes for an entire message to completely arrive at the destination from the time that bit sent out from the source.

Components

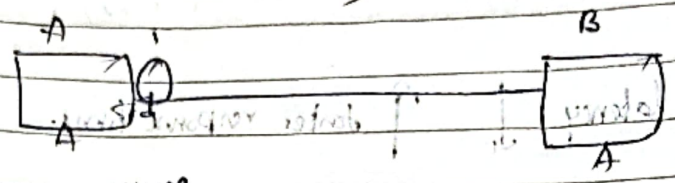
Transmission delay
+
Propagation delay
+
Queuing delay
+
Processing delay

⇒ Latency (+) sum (of all)

Transmission delay:-

Time taken to place the complete data packet on the transmission medium

Transmission →



Message

Placing the message which is ready of transfer on transmission delay

* Sender placing the data which is ready to transfer..

↓

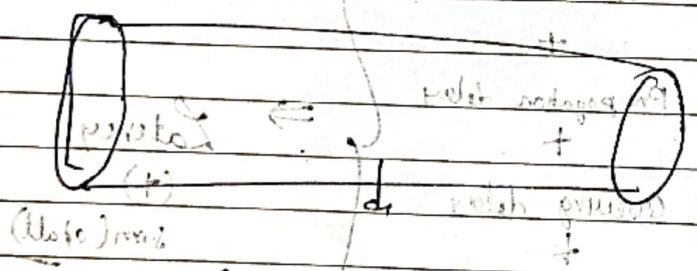
In which there will have a parameter

Bandwidth

↓

It is a measure amount of data capacity that can be transmitted through a short period

In a second how much of amount of data that can be transmitted in the real world.



Size of the channel

transfers

data in per second

100 MBPS

100 MB of data can be transferred in 1 second.
(Per second)

** The More bandwidth the more data can be transferred from the sender to reciever..

$$\text{Transmission time} = \frac{\text{Message size}}{\text{Bandwidth}}$$

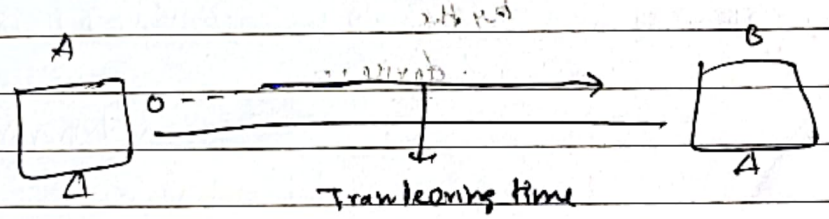
Message size = 1GB ⇔ 1000 MB
Bandwidth = 100 MBPS

$$\text{Transmit time} = \frac{1000}{100} = 10\,ms$$

to transmission ready for transfer

Propagation delay :-

Time it takes for a bit to go from the device A to B

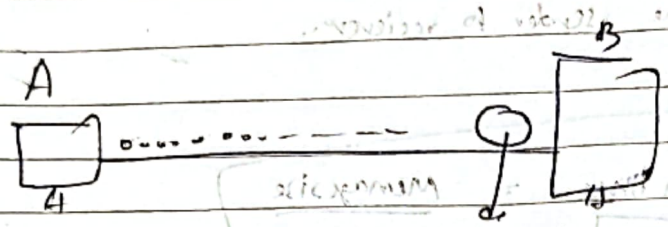


Transferring time taken in a media..

$$\text{propagation} = \frac{\text{distance}}{\text{propagation speed}}$$

Queuing delay

The third component in latency is the queuing time, the time need for each intermediate (or) end device to hold the message before it can processed.



Holding the message before it is processed.

* The queuing time is not fixed factor; changes with load imposed on the network

Processing Delay

* How much time the node takes to process the message
* Time taken process a request which is send by the device.

Throughput

Throughput is the actual amount of data successfully transferred per second..

* Even if a network has high bandwidth, the real data transfer might be lower due to the congestion(or) delay..

* Throughput is the real performance, not a theoritical capacity it is an actual thing made out to an device how much data has accurately taken...

Ex

Bandwidth = 100 Mbps..
Actual transfer = 70 Mbps..
Then
Throughput = 70Mbps

when there is a high through the speed of this will have great

Latency → delay in communication
Bandwidth → Maximum data capacity
Throughput → actual data transfer rate..